Fiscal Year:	FY 2020	**Task Last Updated:**	FY 01/14/2021
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Free Radical Theory of Aging in Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	Ruth.K.Globus@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	08/01/2016	**End Date:**	05/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 5/31/2020 per F. Hernandez/ARC (Ed., 8/12/2020) NOTE: End date changed to 7/31/2020 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: Change in period of performance to 8/1/2016-7/31/2018 (originally 11/1/2014-10/31/2017) per F. Hernandez/ARC (Ed., 7/29/16)		
Key Personnel Changes/Previous PI:	October 2017 report: New Co-Investigators: Candice Tahimic (NASA Ames Research Center); David Goukassian (Temple University). Remove Co-Investigator: Mohammed Shahnazari.		
COI Name (Institution):	Alwood, Joshua Ph.D. (NASA Ames Research Center) Berkowitz, Daniel M.D., Ph.D. (Johns Hopkins School of Medicine) Limoli, Charles Ph.D. (University of California, Irvine) Santhanam, Lakshmi Ph.D. (Johns Hopkins University) Schreurs, Ann-Sofie Ph.D. (NASA Ames Research Center) Tahimic, Candice GT Ph.D. (NASA Ames Research Center) Goukassian, David M.D., Ph.D. (Temple University)		
Grant/Contract No.:	Internal Project		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Many of the adverse physiological effects of spaceflight on astronauts, such as cardiovascular and musculoskeletal deconditioning, resemble those of aging on Earth, leading to the hypothesis that reduced gravity accelerates human aging. This idea is based on numerous spaceflight findings at the organ-system level observed in both humans and animals. Possible specific, mechanistic links between aging and gravity at the molecular level have yet to be established; this is the knowledge gap that we will address in the proposed research project. Our Specific Aims will test responses to long duration spaceflight of a range of tissues (i.e., neural, cardiac, skeletal) in both wild type (control) and transgenic mice. The transgenic mice are engineered to have less oxidative damage in mitochondria, which may occur as a consequence of exposure to long duration spaceflight as well as progressive aging. Female mice (both wildtype and transgenic) will be exposed to spaceflight or kept as ground controls for as long as two months, then tissues (neural, heart, bone) recovered for subsequent analysis of gene expression, oxidative stress biomarkers, and tissue functions. Thus, results from this research will identify basic molecular mechanisms that may underlie degenerative changes arising during prolonged spaceflight or progressive aging on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this research will identify basic molecular mechanisms that may underlie degenerative changes arising during prolonged spaceflight or progressive aging on Earth.
Task Progress:	Ed. NOTE (January 2021): Compiled from report provided to NASA Ames in September 2020; Principal Investigator Ruth Globus, PhD, has retired and CoInvestigator Candice Tahimic, PhD, will be continuing the work in a forthcoming grant. This project was approved to be conducted in two phases: ground-based and flight (Phase 1 and 2, respectively). The aim of the ground-based phase is to conduct Aim 1 of the original flight proposal using a ground-based model for spaceflight (hindlimb unloading, HU). Specifically, we aimed to determine whether spaceflight causes oxidative damage to tissues and accelerates tissue degeneration. During the ground-based phase of this study, we also embarked on making refinements to the standard NASA Ames HU model to facilitate comparison of our rodent flight data with the HU model. Lastly, we tested the feasibility of applying new methodologies to our study with the overall goal of maximizing scientific data from limited samples such as that from spaceflight.
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Articles in Peer-reviewed Journals	Schreurs AS, Torres S, Truong T, Moyer EL, Kumar A, Tahimic CGT, Alwood JS, Globus RK. "Skeletal tissue regulation by catalase over-expression in mitochondria." Am J Physiol Cell Physiol. 2020 Oct 1;319(4):C734-C745. Epub 2020 Aug 12. https://doi.org/10.1152/ajpcell.00068.2020 ; PMID: 32783660 , Oct-2020
Articles in Peer-reviewed Journals	Paul AM, Mhatre SD, Cekanaviciute E, Schreurs A-S, Tahimic CGT, Globus RK, Anand S, Crucian BE, Bhattacharya S. "Neutrophil-to-lymphocyte ratio: A biomarker to monitor the immune status of astronauts." Front Immunol. 2020 Nov 2;11:564950. https://doi.org/10.3389/fimmu.2020.564950 ; PMID: 33224136; PMCID: PMC7667275 , Nov-2020
Articles in Peer-reviewed Journals	Rubinstein L, Schreurs A-S, Torres SM, Steczina S, Lowe MG, Kiffer F, Allen AR, Ronca AE, Sowa MB, Globus RK, Tahimic CGT. "Overexpression of catalase in mitochondria mitigates changes in hippocampal cytokine expression following simulated microgravity and isolation." npj Microgravity. 2021 Jul 6;7(1):24. https://doi.org/10.1038/s41526-021-00152-w ; PMID: 34230490 , Jul-2021